Clustering report

Task 1

Testing result of purity function **compute_purity(y_true, y_pred):**

```
Purity: 75.00%
```

Task 2

Testing result of SSE function **compute_sse(x, y_pred):**

```
My SSE: 78.9451
KMeans SSE (scaled): 78.9451
```

Task 3 K-means on the dataset with k=4

1. Percentage of Data Points Assigned to Each Cluster:

```
Cluster Percentages (%):
0    18.18
1    41.38
2     7.32
3    33.12
```

2. Overall purity: The overall purity of the clustering result is **81.30%**, which suggests that the clusters generally align well with the true air quality labels.
3. Purity of Each cluster:

```
Cluster 0: 59.19%
Cluster 1: 96.52%
Cluster 2: 46.45%
Cluster 3: 82.13%
Cluster with the highest purity: Cluster 1
```

Cluster 1 has the highest purity at 96.52%, indicating that it aligns closely with a single air quality category and is the most reliable cluster.

Task 4 K-Means Clustering with Multiple (k) Values

K-Means Experiment Results:

	k	Purity	SSE
0	2	0.59634	27447.015716
1	3	0.76350	22364.604964
2	4	0.80280	19687.694874
3	10	0.87814	14844.681977
4	20	0.89036	12003.423479
5	30	0.89830	10741.957436

1. Best (k) for purity: The highest purity is achieved at (k=30), with a value of 0.89830 (89.83%). As (k) increases, purity generally improves because more clusters allow finer granularity, which tends to align more closely with the true class labels.
2. Best (k) for SSE: The lowest SSE is observed at (k=30), with a value of 10741.957436. SSE decreases with increasing (k) because smaller clusters minimize the squared distances between points and their respective cluster centers.
3. Change in Purity w.r.t (k):
   Trend: Purity increases steadily as (k) increases. This is expected, as higher values of (k) result in more clusters, which can better capture variations in the data, leading to clusters that align more closely with the true labels. As (k) grows, clusters tend to overfit the data, capturing even small variations. This can improve purity at the expense of generalizability.
4. Change in SSE w.r.t (k):
   Trend: SSE decreases consistently with increasing (k). This is because more clusters reduce the average distance between points and their cluster centers. With more clusters, each cluster covers a smaller portion of the data, leading to tighter groupings and lower SSE values. However, excessive clusters may lead to over-segmentation and reduced interpretability.

Task 5 DBSCAN Clustering Experiment

DBSCAN Experiment Results:

	eps	Number of Clusters	Number of Anomalies	Purity	SSE	Silhouette Coefficient
0	0.8	12	2656	0.790102	6850.021544	-0.009287
1	0.9	11	2071	0.666439	10580.821406	0.007289
2	1.0	8	1532	0.571223	15668.087223	0.100165
3	1.1	10	1136	0.520963	19746.073374	0.097361
4	1.2	4	854	0.482393	23556.197082	0.227144

1. Best (eps) for Purity: The highest purity is observed at (eps=0.8), with a value of 0.790102 (79.01%). Smaller eps values tend to create more precise clusters by preventing excessive merging.
2. Best (eps) for SSE: The lowest SSE is achieved at (eps=0.8), with a value of 6850.021544. This is expected, as tighter clusters reduce the squared distances between points and their cluster centers.
3. Best (eps) for Silhouette Coefficient: The highest silhouette coefficient is observed at (eps=1.2), with a value of 0.227144. Larger eps values create fewer, more distinct clusters, improving separation but potentially reducing cohesion.

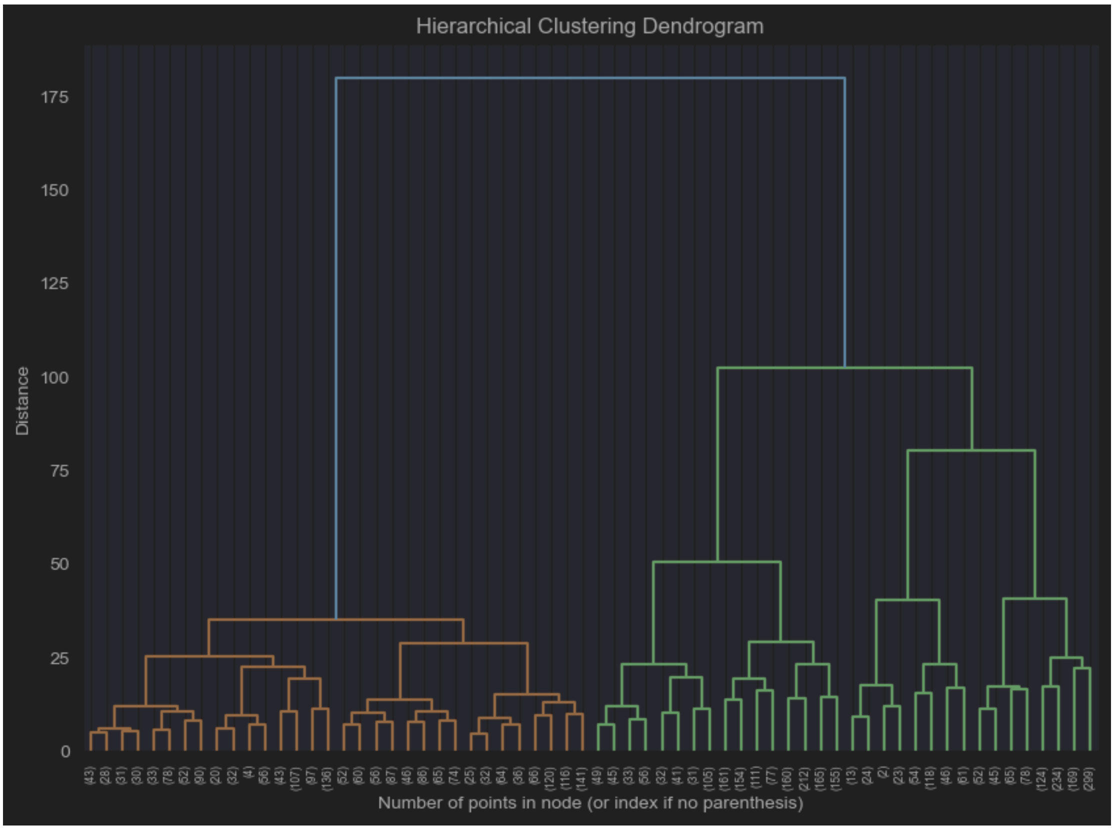
How Metrics Change with (eps):

Purity decreases as eps increases, reflecting larger clusters that merge diverse points, reducing alignment with true class labels.

SSE increases with eps, as larger clusters encompass points farther from their centers, increasing the sum of squared errors.

The silhouette coefficient improves with higher eps, indicating better separation between clusters. However, negative or near-zero values (e.g., (eps=0.8, 0.9)) suggest overlapping clusters.

Task 6: Hierarchical Clustering Experiment



Hierarchical Clustering Results:

	Distance Threshold	Purity	SSE	Number of Clusters
0	25	0.8818	15675.166567	11
1	75	0.8450	20364.869108	4
2	125	0.6940	28813.695008	2

Threshold for 5 Clusters: Based on the dendrogram, a distance_threshold of 46 can be chosen to detect 5 clusters. At this threshold, the splits align with the desired number of clusters, ensuring a balance between the number of clusters and their interpretability.

Best Threshold for Purity: The highest purity ((0.8818)) is achieved at (distance_threshold=25), where 11 clusters were formed. More clusters generally allow for finer alignment with the true labels.

Best Threshold for SSE: The lowest SSE ((15675.166567)) also occurs at (distance_threshold=25), as smaller and more numerous clusters minimize the squared errors.

Trade-Off: As the distance_threshold increases: Purity decreases because larger clusters combine diverse data points, reducing alignment with true labels. SSE increases due to larger clusters covering more dispersed points.

The choice of threshold depends on the application: higher purity and lower SSE for finegrained clustering (e.g., (25)), or fewer, broader clusters for simplicity (e.g., (75)).